



# NORMANHURST BOYS HIGH SCHOOL

## MATHEMATICS ADVANCED

Year 11 Course Assessment Task 3

Wednesday September 18, 2019

### General instructions

- Working time – 2 hours  
(plus 10 minutes reading time)
- Write your answers in the space provided.
- Write using a blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, bundle any additional sheets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

☐ 11MAT.6 – Dr Lawson

NESA STUDENT #: .....

Marker's use only.

| QUESTION | 1-10            | 11              | 12              | 13              | 14              | 15              | Total           | % |
|----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|---|
| MARKS    | $\overline{10}$ | $\overline{18}$ | $\overline{10}$ | $\overline{16}$ | $\overline{16}$ | $\overline{13}$ | $\overline{83}$ |   |

## Section I

10 marks

Attempt Question 1 to 10

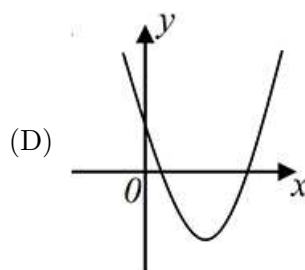
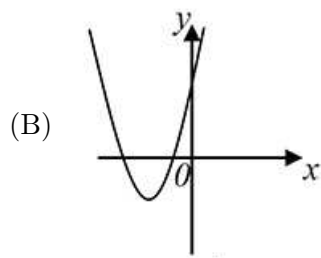
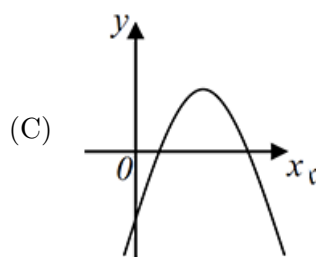
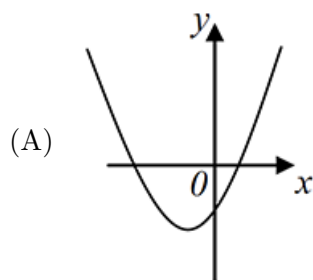
Allow approximately 10 minutes for this section

Mark your answers on the answer grid provided (labelled as page 18).

### Questions

### Marks

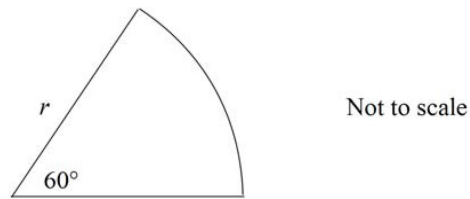
- When the denominator is rationalised, what is  $\frac{1}{\sqrt{5} - \sqrt{3}}$  equal to? 1  
 (A)  $\frac{\sqrt{5} - \sqrt{3}}{2}$       (B)  $\frac{\sqrt{5} - \sqrt{3}}{16}$       (C)  $\frac{\sqrt{5} + \sqrt{3}}{2}$       (D)  $\sqrt{5} + \sqrt{3}$
- Which of the following parabolas could have the equation  $y = (x - 5)(x - 1)$ ? 1



- What is range of the function  $f(x) = \sqrt{4 - x^2}$ ? 1  
 (A)  $(0, 4)$       (B)  $[-4, 4]$       (C)  $(-2, 2)$       (D)  $[0, 2]$
- What is  $e^{2 \log_e x}$  is equal to? 1  
 (A)  $\sqrt{x}$       (B)  $x^2$       (C)  $2x$       (D)  $e^2 x$
- If  $4^{x-1} = 32$ , what is the value of  $x$ ? 1  
 (A) 10      (B) 3.5      (C) 3      (D) 6

6. The sector below has an area of  $20\pi$  square units.

1

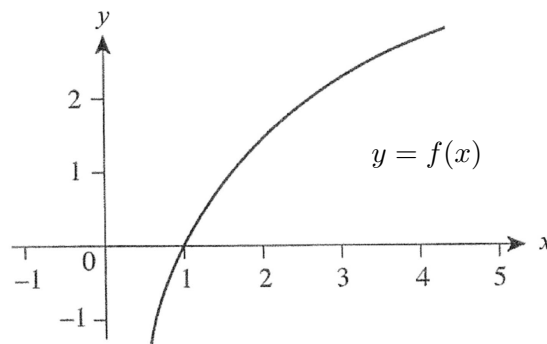


What is the value of  $r$ ?

- (A)  $\sqrt{120}$       (B)  $\sqrt{120}\pi$       (C)  $\sqrt{\frac{\pi}{3}}$       (D)  $\sqrt{\frac{1}{3}}$

7. The graph of  $y = f(x)$  below has which of the following properties?

1



- (A)  $f'(x) > 0$  and  $f''(x) > 0$       (C)  $f'(x) < 0$  and  $f''(x) > 0$   
 (B)  $f'(x) > 0$  and  $f''(x) < 0$       (D)  $f'(x) < 0$  and  $f''(x) < 0$

8. Evaluate  $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$

1

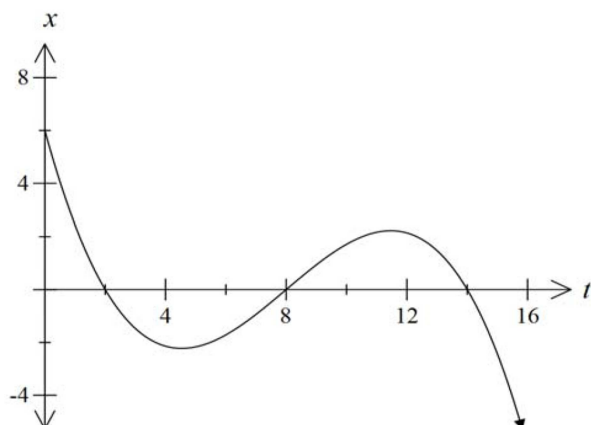
- (A)  $\infty$       (B) 0      (C) 4      (D) 2

9. If  $f(x) = a^x + a^{-x}$ , which of the following expressions is equal to  $[f(x)]^2$ ?

1

- (A)  $1 - f(2x)$       (B)  $1 + f(2x)$       (C)  $2 - f(2x)$       (D)  $2 + f(2x)$

10. The displacement,  $x$  metres, from the origin of a particle moving in a straight line at any time ( $t$  seconds) is shown in the graph. 1



When was the particle first at rest?

- (A)  $t = 0$                       (B)  $t = 2$                       (C)  $t = 4.5$                       (D) Never

**Examination continues overleaf...**

## Section II

70 marks

Attempt Questions 11 to 15

Allow approximately 1 hour and 50 minutes for this section.

Question 11 (18 Marks)

Marks

(a) Solve for  $x$ :  $\frac{x+5}{3} - 2x = 7$  **2**

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(b) Factorise  $3x^2 + 5x - 2$ . **2**

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(c) Simplify  $\frac{a^2 \times a^{x-4}}{a^{1-2x}}$  **2**

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(d) Consider the function  $f(x) = \frac{3}{x+1}$ . **1**

i. Show that  $f(x)$  is neither an even nor an odd function.

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ii. State the domain of  $y = f(x)$  in interval notation. **1**

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(e) Given

$$f(x) = \begin{cases} -4 & x \leq -2 \\ x & -2 < x \leq 1 \\ x^2 - 1 & x > 1 \end{cases}$$

i. Find  $f(-1)$ . **1**

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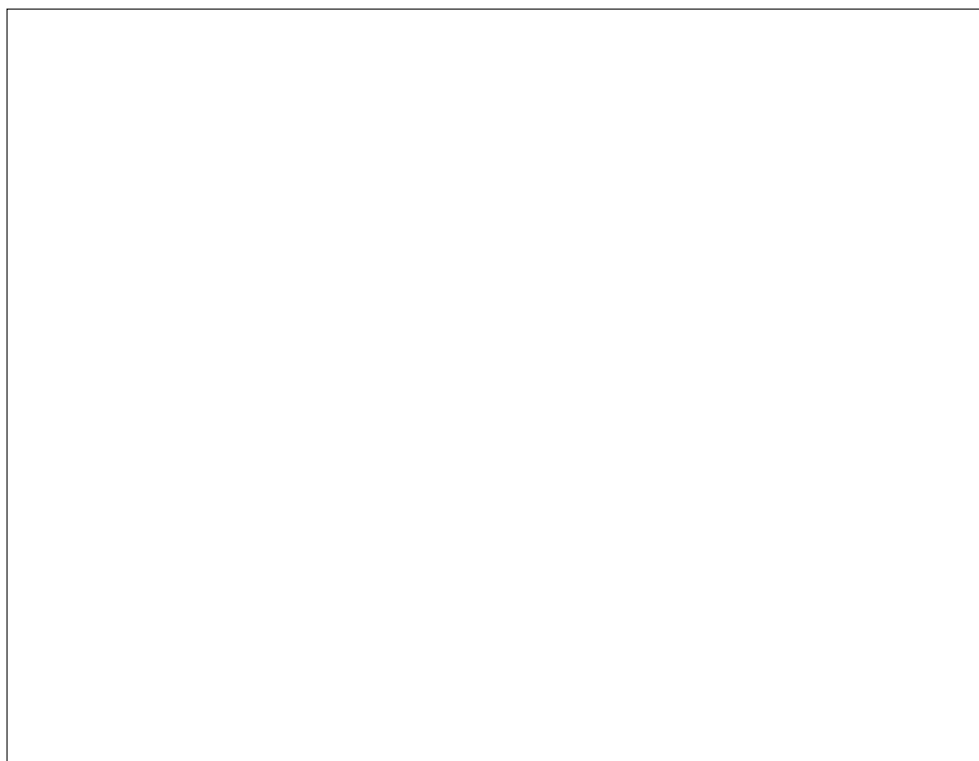
ii. Find  $f(a^2 + 1)$ . Simplify your answer. **2**

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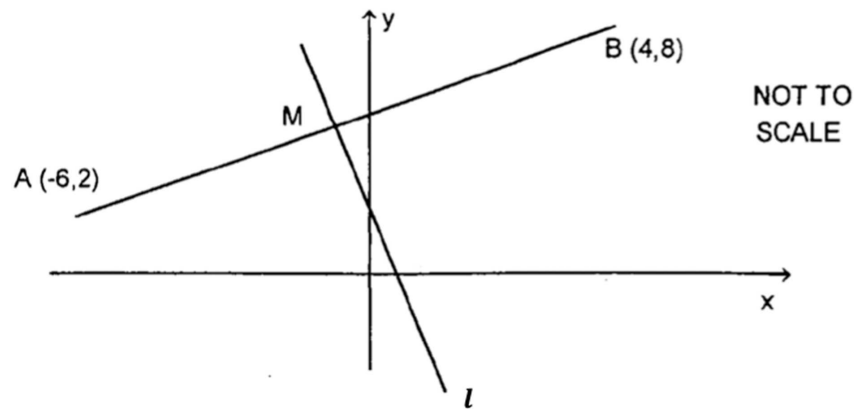
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iii. Sketch the graph of  $y = f(x)$  **3**

- (f) The diagram below show  $A(-6, 2)$  and  $B(4, 8)$ .



- i. Find  $M$ , the midpoint of  $AB$ . 1

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- ii. Show that the gradient of  $AB$  is  $\frac{3}{5}$ . 1

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- iii. Line  $l$  is the perpendicular bisector of  $AB$ . Show that the equation of line  $l$  is  $5x + 3y - 10 = 0$ . 2

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**Question 12** (10 Marks)**Marks**

(a) Find the derivative of the followings, leaving your answer in simplest form.

i.  $y = -3x^4$  **1**

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ii.  $f(x) = (3x^2 - 5)^6$  **2**

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iii.  $f(x) = \frac{2x - 1}{x + 4}$  **2**

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iv.  $y = 3x^2\sqrt{x-2}$ . (No need to fully simplify) **2**

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(b) Differentiate  $f(x) = x^2$  from first principles. **2**

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- (c) Find the  $x$  coordinate of the point on the curve  $y = x^2 + 5$  where the tangent has the gradient of  $-2$ .

**1**

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**Examination continues overleaf...**

**Question 13** (16 Marks)**Marks**

- (a) Show that  $f(x) = \frac{1}{x^3} - x^3$  is always decreasing. **2**

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- (b) Consider  $f(x) = x^3 + x^2 - x - 1$ .
- i. Given that  $x^3 + x^2 - x - 1 = (x - 1)(x + 1)^2$ , find the  $x$  and  $y$ -intercepts. **2**

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- ii. Find the stationary points and determine their nature. **4**

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- iii. Find the point of inflexion.

**2**

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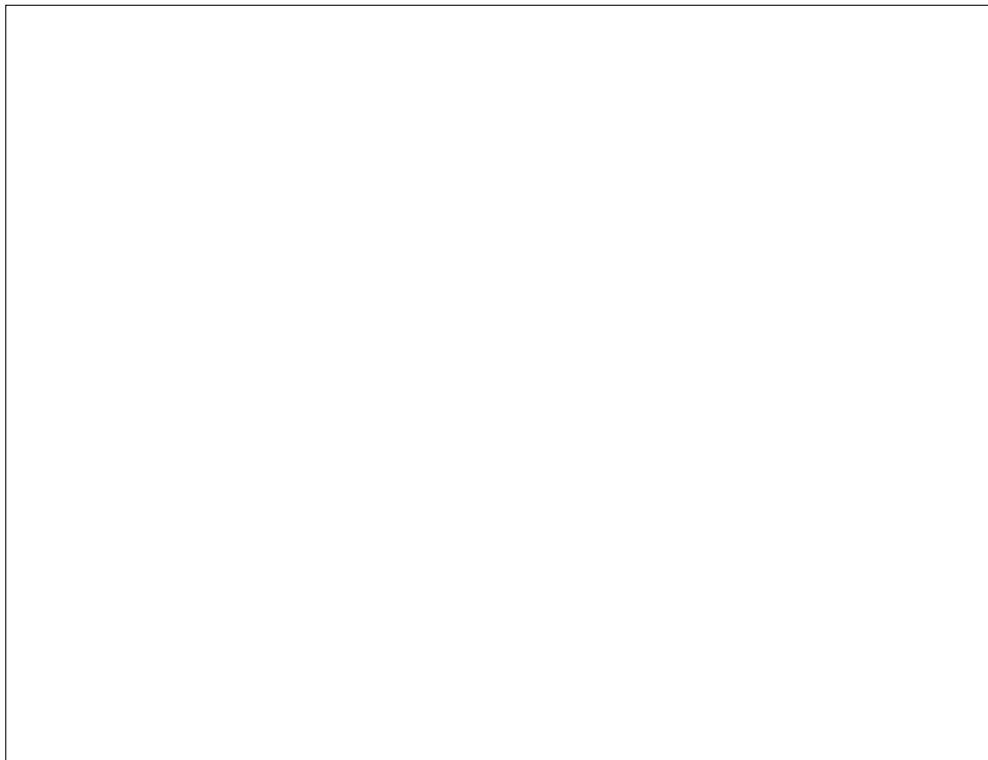
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- iv. Sketch  $y = x^3 + x^2 - x - 1$ , showing all important features.

**2**

- v. How many solutions does the equation  $x^3 + x^2 - x - 1 = 1$  have?

**1**

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- (c) The volume, in litres, of water in a bathtub,  $t$  minutes after it has started to drain, is given by the equation

$$V = 50(14 - t)^2$$

- i. Calculate the initial volume of water in the bathtub. **1**

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- ii. At what rate does the water drain out of the bathtub after 6 minutes? **2**

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**Examination continues overleaf...**

**Question 14** (16 Marks)**Marks**

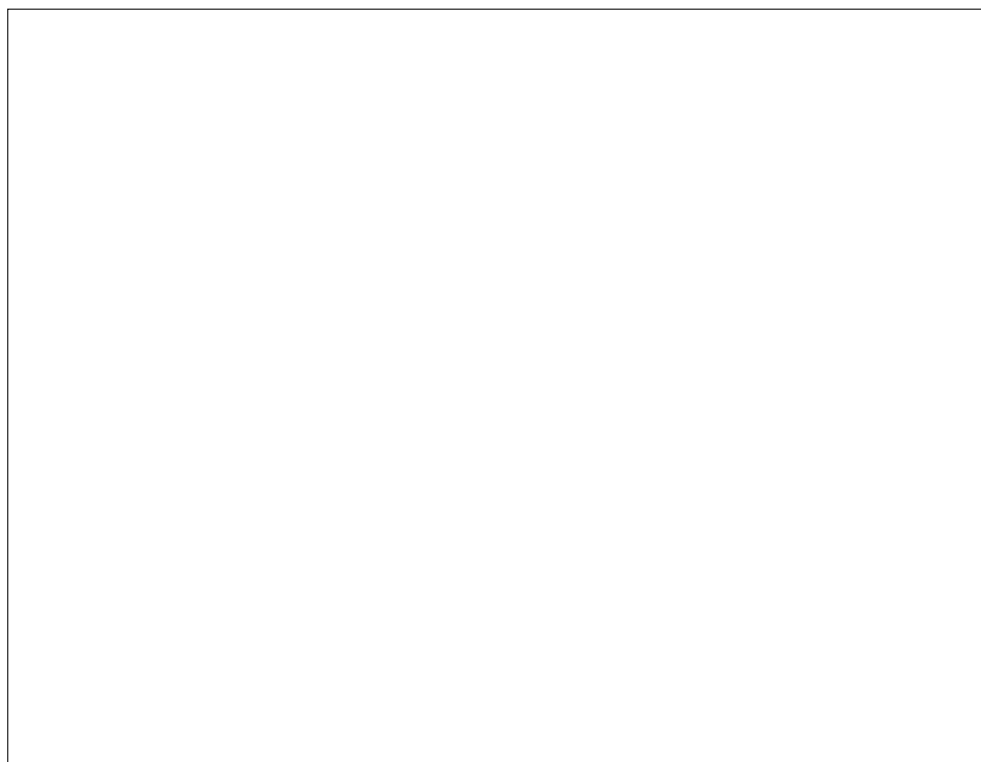
- (a) The angle of a sector in a circle of radius 9 cm is  $\frac{\pi}{5}$  radians. Find the exact value of the perimeter of the sector. **2**

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- (b) i. On the same number plane, graph  $y = \cos x$  and  $y = \sin 2x$  for  $0 \leq x \leq \pi$ , showing all intercepts. **2**



- ii. Show that the curves first intersect at  $x = \frac{\pi}{6}$ . **1**

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- (c) Show that the exact value of  $\cos \frac{\pi}{4} + \sin \frac{2\pi}{3}$  is  $\frac{\sqrt{2} + \sqrt{3}}{2}$  **1**

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- (d) i. Show that

**2**

$$\frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta = \tan \theta$$

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- ii. Hence, or otherwise, solve

**2**

$$\frac{\sin^3 \theta}{\cos \theta} = 1 - \sin \theta \cos \theta$$

where  $0 \leq x \leq 2\pi$ .

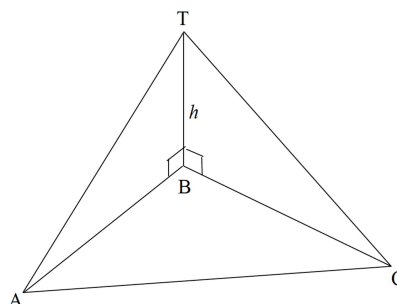
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- (e) The angle of elevation of a vertical tower  $BT$  of height  $h$  metres, at a point  $A$  due west of it, is  $14^\circ$ . From the point  $C$ , the bearing of the tower is  $343^\circ$  and the angle of elevation is  $11^\circ$ . The points  $A$  and  $C$  are 1 000 m apart and on the same level as the base  $B$  of the tower.



- i. By drawing a diagram showing the bearings, explain why  $\angle ABC = 107^\circ$ . 1

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- ii. Show that  $AB = h \tan 76^\circ$  1

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- iii. Find a similar expression for  $CB$ . 1

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- iv. Show that 2

$$h^2 = \frac{1000^2}{\tan^2 76^\circ + \tan^2 79^\circ - 2 \tan 76^\circ \tan 79^\circ \cos 107^\circ}$$

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- v. Hence, find the height of the tower, correct to the nearest metre. 1

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**Examination continues overleaf...**

**Question 15** (13 Marks)**Marks**

- (a) Solve the equation
- $2 \log_e(x - 5) = \log_e(2x - 7)$

**3**

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- (b) Differentiate with respect to
- $x$
- :

i.  $y = e^{7x}$

**1**

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ii.  $y = \frac{e^x}{x^2}$

**2**

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- (c) Find the equation of the normal to the curve
- $y = e^x(x^2 + 1)$
- at
- $(0, 1)$
- .

**3**

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- (d) i. Solve for  $x$ :  $x^2 + 2x + 1 = 0$ . **1**

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- ii. The displacement of an object at time ( $t$ ) seconds is given by **3**

$$x = e^{-2t} + 4e^{-t} - 2t$$

Hence, or otherwise, show that the object never comes to rest.  
(Hint: Let  $m = e^{-t}$ )

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**End of paper.**

## Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

NESA STUDENT #: .....

Class (please ✓)

### Directions for multiple choice answers

- Read each question and its suggested answers.
- Select the alternative (A), (B), (C), or (D) that best answers the question.
- Mark only one circle per question. There is only *one* correct choice per question.
- Fill in the response circle completely, using blue or black pen, e.g.

(A) (B) ● (D)

- If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) (B) ~~●~~ ●

- If you continue to change your mind, write the word **correct** and clearly indicate your final choice with an arrow as shown below:

(A) (B) ~~●~~ ~~●~~ <sup>correct</sup>

1 – (A) (B) (C) (D)

2 – (A) (B) (C) (D)

3 – (A) (B) (C) (D)

4 – (A) (B) (C) (D)

5 – (A) (B) (C) (D)

6 – (A) (B) (C) (D)

7 – (A) (B) (C) (D)

8 – (A) (B) (C) (D)

9 – (A) (B) (C) (D)

10 – (A) (B) (C) (D)



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☐ 11MAT.6 – Dr Lawson

**NAME** .....

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| QUESTION | 1-10            | 11              | 12              | 13              | 14              | 15              | Total           | % |
|----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|---|
| MARKS    | $\overline{10}$ | $\overline{18}$ | $\overline{10}$ | $\overline{16}$ | $\overline{16}$ | $\overline{13}$ | $\overline{83}$ |   |

## Section I

10 marks

Attempt Question 1 to 10

Allow approximately 10 minutes for this section

Mark your answers on the answer grid provided (labelled as page 18).

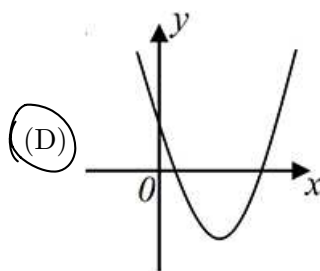
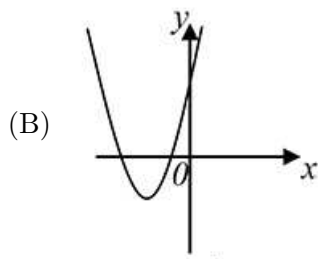
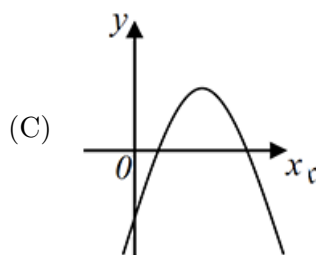
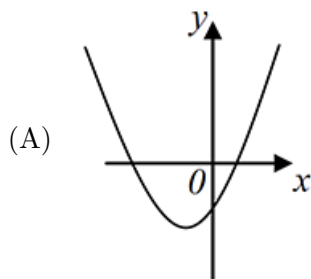
### Questions

### Marks

1. When the denominator is rationalised, what is  $\frac{1}{\sqrt{5}-\sqrt{3}}$  equal to? 1

(A)  $\frac{\sqrt{5}-\sqrt{3}}{2}$       (B)  $\frac{\sqrt{5}-\sqrt{3}}{16}$       (C)  $\frac{\sqrt{5}+\sqrt{3}}{2}$       (D)  $\sqrt{5}+\sqrt{3}$

2. Which of the following parabolas could have the equation  $y = (x-5)(x-1)$ ? 1



3. What is range of the function  $f(x) = \sqrt{4-x^2}$ ? 1

(A)  $(0, 4)$       (B)  $[-4, 4]$       (C)  $(-2, 2)$       (D)  $[0, 2]$

4. What is  $e^{2\log_e x}$  is equal to? 1

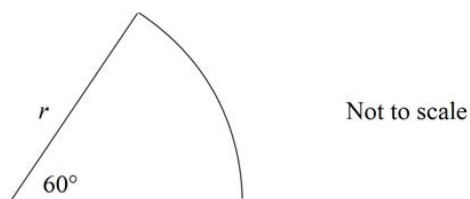
(A)  $\sqrt{x}$       (B)  $x^2$       (C)  $2x$       (D)  $e^2x$

5. If  $4^{x-1} = 32$ , what is the value of  $x$ ? 1

(A) 10      (B) 3.5      (C) 3      (D) 6

6. The sector below has an area of  $20\pi$  square units.

1

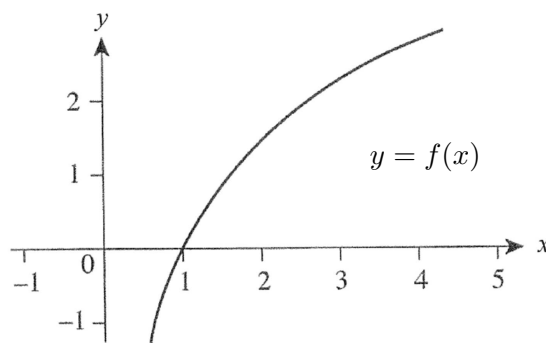


What is the value of  $r$ ?

- (A)  $\sqrt{120}$  (B)  $\sqrt{120\pi}$  (C)  $\sqrt{\frac{\pi}{3}}$  (D)  $\sqrt{\frac{1}{3}}$

7. The graph of  $y = f(x)$  below has which of the following properties?

1



- (A)  $f'(x) > 0$  and  $f''(x) > 0$  (C)  $f'(x) < 0$  and  $f''(x) > 0$

- (B)  $f'(x) > 0$  and  $f''(x) < 0$  (D)  $f'(x) < 0$  and  $f''(x) < 0$

8. Evaluate  $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$

1

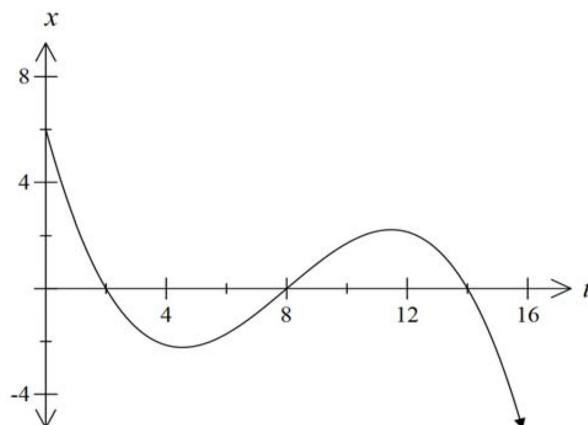
- (A)  $\infty$  (B) 0 (C) 4 (D) 2

9. If  $f(x) = a^x + a^{-x}$ , which of the following expressions is equal to  $[f(x)]^2$ ?

1

- (A)  $1 - f(2x)$  (B)  $1 + f(2x)$  (C)  $2 - f(2x)$  (D)  $2 + f(2x)$

10. The displacement,  $x$  metres, from the origin of a particle moving in a straight line at any time ( $t$  seconds) is shown in the graph. 1



When was the particle first at rest?

- (A)  $t = 0$       (B)  $t = 2$       (C)  $t = 4.5$       (D) Never

**Examination continues overleaf...**

## Section II

70 marks

Attempt Questions 11 to 15

Allow approximately 1 hour and 50 minutes for this section.

### Question 11 (18 Marks)

Marks

- (a) Solve for  $x$ :  $\frac{x+5}{3} - 2x = 7$  2

$$x+5-6x=21$$

$$-5x=16$$

$$\therefore x = \frac{-16}{5}$$

- (b) Factorise  $3x^2 + 5x - 2$ . 2

$$\begin{array}{cc} 1 & 2 \\ 3 & -1 \end{array}$$

$$= (x+2)(3x-1)$$

- (c) Simplify  $\frac{a^2 \times a^{x-4}}{a^{1-2x}}$  2

$$= \frac{a^{x-2}}{a^{1-2x}}$$

$$= a^{x-2-(1-2x)}$$

$$= a^{x-2-1+2x}$$

$$= a^{3x-3}$$

- (d) Consider the function  $f(x) = \frac{3}{x+1}$ . 1

- i. Show that  $f(x)$  is neither an even nor an odd function. 1

$$f(-x) = \frac{3}{-x+1}$$

$$= -\frac{3}{x-1}$$

$$\neq -f(x) \quad \therefore \text{neither even nor odd}$$

- ii. State the domain of  $y = f(x)$  in interval notation. 1

$$x \in (-\infty, -1) \cup (-1, \infty)$$

(e) Given

$$f(x) = \begin{cases} -4 & x \leq -2 \\ x & -2 < x \leq 1 \\ x^2 - 1 & x > 1 \end{cases}$$

i. Find  $f(-1)$ .

1

$$f(-1) = -1$$

ii. Find  $f(a^2 + 1)$ . Simplify your answer.

2

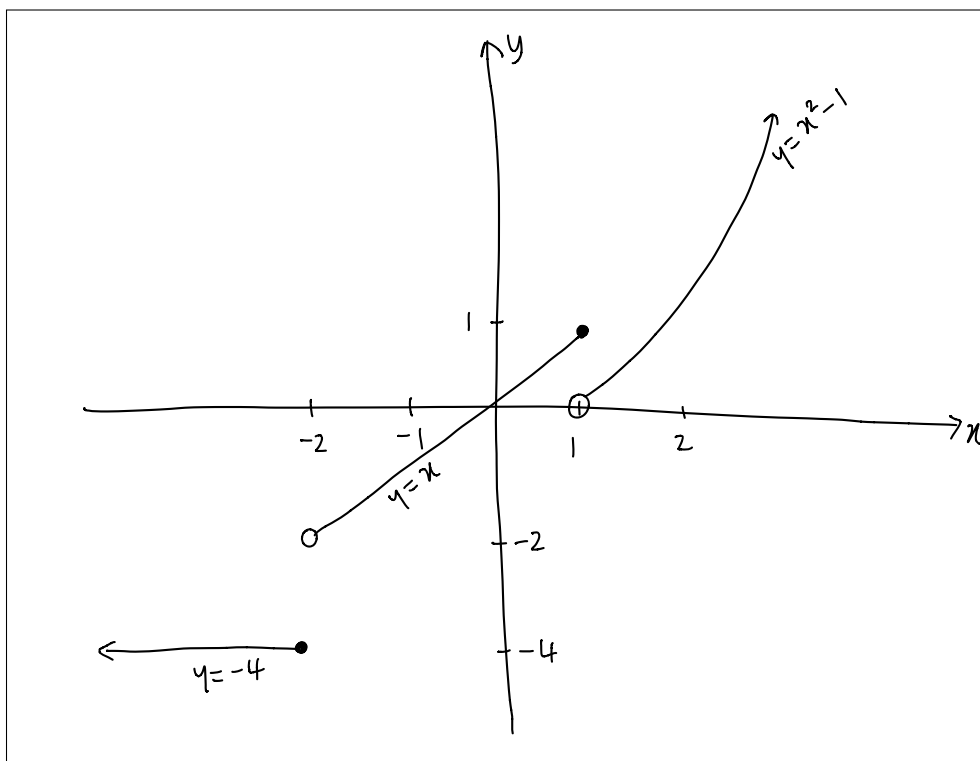
$$f(a^2 + 1) = (a^2 + 1)^2 - 1$$

$$= a^4 + 2a^2 + 1 - 1$$

$$= a^4 + 2a^2$$

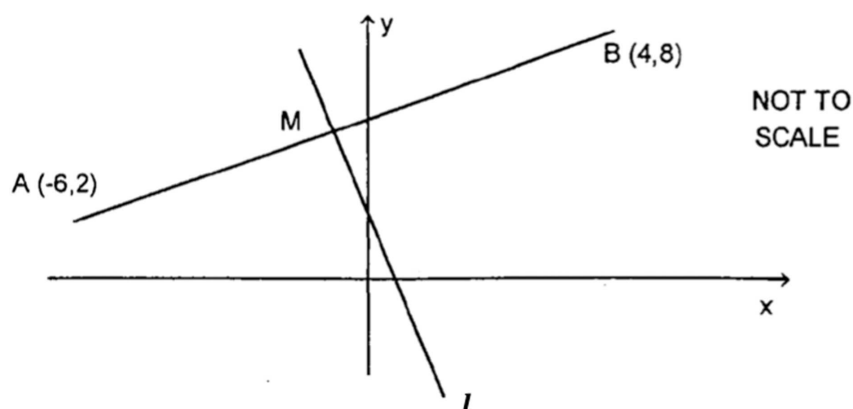
iii. Sketch the graph of  $y = f(x)$ 

3





- (f) The diagram below show  $A(-6, 2)$  and  $B(4, 8)$ .



- i. Find  $M$ , the midpoint of  $AB$ .

1

$$M\left(\frac{-2}{2}, \frac{10}{2}\right)$$

$$= M(-1, 5)$$

- ii. Show that the gradient of  $AB$  is  $\frac{3}{5}$ .

1

$$m_{AB} = \frac{8-2}{4-(-6)}$$

$$= \frac{6}{10} = \frac{3}{5}$$

- iii. Line  $l$  is the perpendicular bisector of  $AB$ . Show that the equation of line  $l$  is  $5x + 3y - 10 = 0$ .

2

$$m_l = -\frac{5}{3}, \quad (-1, 5)$$

$$\therefore \text{Eqn: } y - 5 = -\frac{5}{3}(x + 1)$$

$$3y - 15 = -5x - 5$$

$$5x + 3y - 10 = 0$$

**Question 12** (10 Marks)**Marks**

(a) Find the derivative of the followings, leaving your answer in simplest form.

i.  $y = -3x^4$

**1**

$$y' = -12x^3$$

ii.  $f(x) = (3x^2 - 5)^6$

**2**

$$f'(x) = 6(3x^2 - 5)^5 \cdot 6x$$

$$= 36x(3x^2 - 5)^5$$

iii.  $f(x) = \frac{2x - 1}{x + 4}$

**2**

$$f'(x) = \frac{2(x+4) - (2x-1)}{(x+4)^2}$$

$$= \frac{2x+8-2x+1}{(x+4)^2}$$

$$= \frac{9}{(x+4)^2}$$

iv.  $y = 3x^2\sqrt{x-2}$ . (No need to fully simplify)

**2**

$$= 3x^2(x-2)^{1/2}$$

$$y' = 6x(x-2)^{1/2} + 3x^2 \cdot \frac{1}{2}(x-2)^{-1/2}$$

$$= 6x\sqrt{x-2} + \frac{3x^2}{2\sqrt{x-2}}$$

(b) Differentiate  $f(x) = x^2$  from first principles.**2**

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$

$$= \lim_{h \rightarrow 0} 2x + h$$

$$= 2x$$

- (c) Find the  $x$  coordinate of the point on the curve  $y = x^2 + 5$  where the tangent has the gradient of  $-2$ . 1

$$y' = 2x$$

$$\therefore 2x = -2$$

$$x = -1$$

**Examination continues overleaf...**

**Question 13** (16 Marks)**Marks**

- (a) Show that  $f(x) = \frac{1}{x^3} - x^3$  is always decreasing. 2

$$f(x) = x^{-3} - x^3$$

$$f'(x) = -3x^{-4} - 3x^2$$

$$= -\frac{3}{x^4} - 3x^2$$

For all real  $x$ ,  $x^2 \geq 0$  &  $x^4 \geq 0$

$$\therefore -\frac{3}{x^4} - 3x^2 < 0 \text{ for all } x.$$

$\therefore$  always decreasing

- (b) Consider  $f(x) = x^3 + x^2 - x - 1$ . 2
- i. Given that  $x^3 + x^2 - x - 1 = (x-1)(x+1)^2$ , find the  $x$  and  $y$ -intercepts.

$$x\text{-int} : x = 1, -1$$

$$y\text{-int} : y = -1$$

- ii. Find the stationary points and determine their nature. 4

$$f'(x) = 3x^2 + 2x - 1$$

$$\frac{1}{3} \quad -1$$

$$= (x+1)(3x-1)$$

$$\therefore x = -1, \frac{1}{3}$$

$$y = 0, -\frac{32}{27} \quad \therefore (-1, 0) \text{ \& \& } (\frac{1}{3}, -\frac{32}{27})$$

| $x$  | -2 | -1 | 0  | $\frac{1}{3}$ | 1 |
|------|----|----|----|---------------|---|
| $y'$ | 7  | 0  | -1 | 0             | 4 |
|      | /  | —  | \  | —             | / |

$\therefore$  max @  $(-1, 0)$

min @  $(\frac{1}{3}, -\frac{32}{27})$

- iii. Find the point of inflexion.

2

$$f''(x) = 6x + 2$$

$$6x + 2 = 0$$

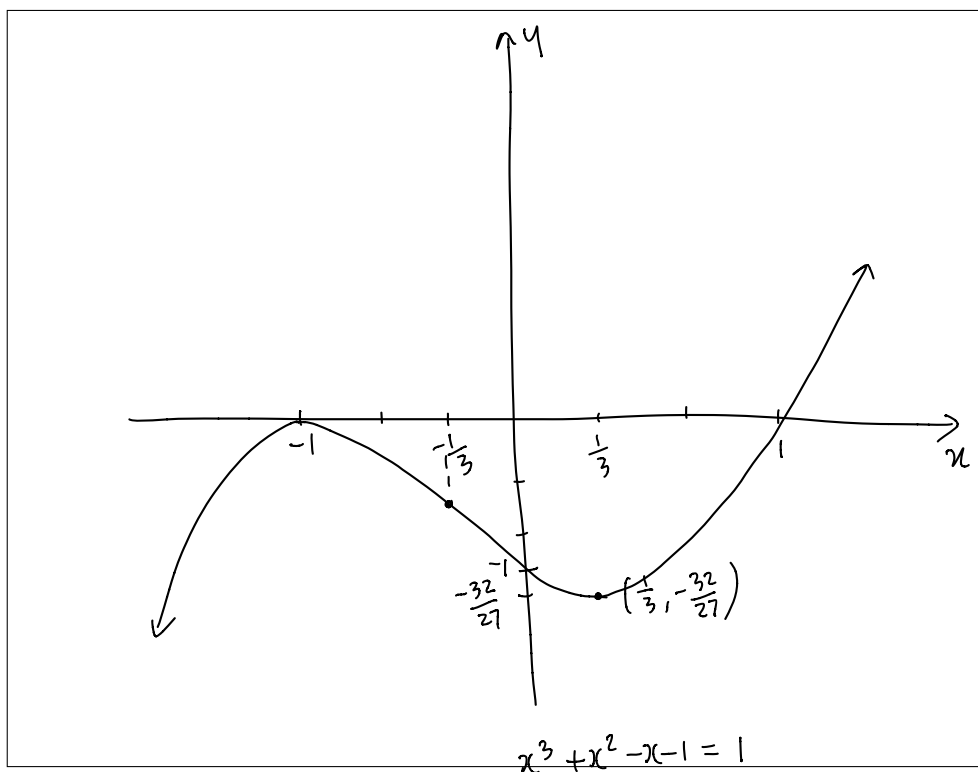
$$x = -\frac{1}{3}, \quad y = -\frac{16}{27}$$

|       |    |                |   |
|-------|----|----------------|---|
| $x$   | -1 | $-\frac{1}{3}$ | 0 |
| $y''$ | -4 | 0              | 2 |

∴ pt of inflexion @  $(-\frac{1}{3}, -\frac{16}{27})$

- iv. Sketch
- $y = x^3 + x^2 - x - 1$
- , showing all important features.

2



- v. How many solutions does the equation
- $x^3 + x^2 - x - 1 = 0$
- have?

1

1 solution

- (c) The volume, in litres, of water in a bathtub,  $t$  minutes after it has started to drain, is given by the equation

$$V = 50(14 - t)^2$$

- i. Calculate the initial volume of water in the bathtub. **1**

When  $t=0$ ,  $V = 50 \times 14^2$

$= 9800 \text{ L}$

- ii. At what rate does the water drain out of the bathtub after 6 minutes? **2**

$\frac{dV}{dt} = 100(14-t)^1 \times (-1)$

$= -100(14-t)$

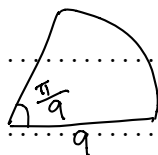
When  $t=6$ ,  $\frac{dV}{dt} = -800 \text{ L/min}$

$\therefore$  The water drains out at  $800 \text{ L/min}$ .

**Examination continues overleaf...**

**Question 14** (16 Marks)**Marks**

- (a) The angle of a sector in a circle of radius 9 cm is  $\frac{\pi}{5}$  radians. Find the exact value of the perimeter of the sector. 2



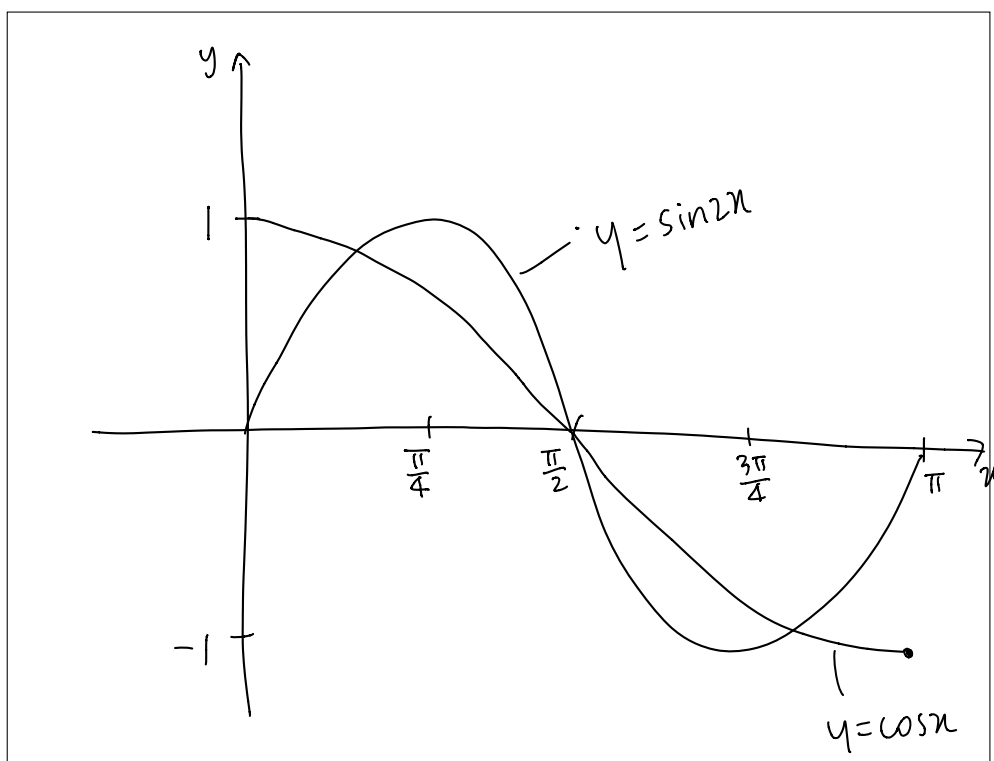
$$l = r\theta = 9 \times \frac{\pi}{5} = \frac{9\pi}{5}$$

$$\therefore \left( \frac{9\pi}{5} + 18 \right) \text{ cm}$$

- (b) i. On the same number plane, graph  $y = \cos x$  and  $y = \sin 2x$  for  $0 \leq x \leq \pi$ , showing all intercepts. 2

$$T = 2\pi$$

$$T = \pi$$



- ii. Show that the curves first intersect at  $x = \frac{\pi}{6}$ . 1

$$\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

$$\sin 2\left(\frac{\pi}{6}\right) = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} \quad \therefore \text{Intersect @ } x = \frac{\pi}{6}$$

- (c) Show that the exact value of  $\cos \frac{\pi}{4} + \sin \frac{2\pi}{3}$  is  $\frac{\sqrt{2} + \sqrt{3}}{2}$  1

$$\cos \frac{\pi}{4} + \sin \frac{2\pi}{3} = \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{2} + \sqrt{3}}{2}$$

- (d) i. Show that

2

$$\frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta = \tan \theta$$

$$\text{LHS} = \frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta$$

$$= \frac{\sin^3 \theta + \sin \theta \cos^2 \theta}{\cos \theta}$$

$$= \frac{\sin \theta (\sin^2 \theta + \cos^2 \theta)}{\cos \theta}$$

$$= \tan \theta$$

$$= \text{RHS}$$

- ii. Hence, or otherwise, solve

2

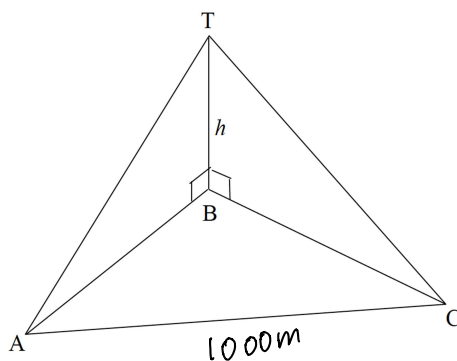
$$\frac{\sin^3 \theta}{\cos \theta} = 1 - \sin \theta \cos \theta$$

where  $0 \leq \theta \leq 2\pi$ .

$$\tan \theta - \sin \theta \cos \theta = 1 - \sin \theta \cos \theta$$

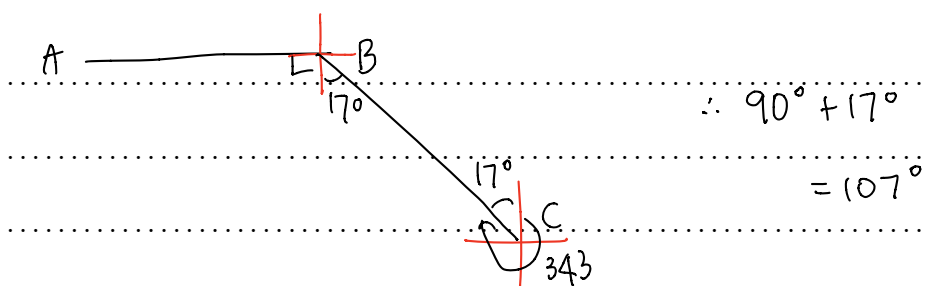
$$\tan \theta = 1, \quad \theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

- (e) The angle of elevation of a vertical tower
- $BT$
- of height
- $h$
- metres, at a point
- $A$
- due west of it, is
- $14^\circ$
- . From the point
- $C$
- , the bearing of the tower is
- $343^\circ$
- and the angle of elevation is
- $11^\circ$
- . The points
- $A$
- and
- $C$
- are 1 000 m apart and on the same level as the base
- $B$
- of the tower.

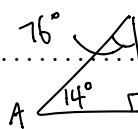




- i. By drawing a diagram showing the bearings, explain why  $\angle ABC = 107^\circ$ . 1



- ii. Show that  $AB = h \tan 76^\circ$  1



$$\tan 76^\circ = \frac{AB}{h}$$

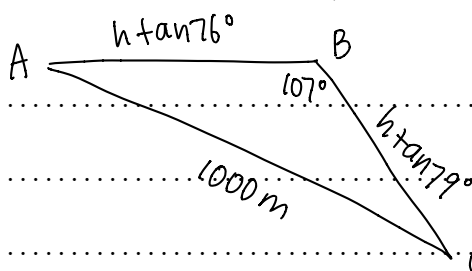
$$\therefore AB = h \tan 76^\circ$$

- iii. Find a similar expression for  $CB$ . 1

$$CB = h \tan 79^\circ$$

- iv. Show that 2

$$h^2 = \frac{1000^2}{\tan^2 76^\circ + \tan^2 79^\circ - 2 \tan 76^\circ \tan 79^\circ \cos 107^\circ}$$



cosine rule

$$1000^2 = h^2 \tan^2 76^\circ + h^2 \tan^2 79^\circ - 2 \times h \tan 76^\circ \times h \tan 79^\circ \times \cos 107^\circ$$

$$1000^2 = h^2 (\tan^2 76^\circ + \tan^2 79^\circ - 2 \tan 76^\circ \tan 79^\circ \cos 107^\circ)$$

$$\therefore h^2 = \frac{1000^2}{\tan^2 76^\circ + \tan^2 79^\circ - 2 \tan 76^\circ \tan 79^\circ \cos 107^\circ}$$

- v. Hence, find the height of the tower, correct to the nearest metre. 1

$$h = 135.310 \dots = 135 \text{ m (to the nearest m)}$$

Examination continues overleaf...

**Question 15 (13 Marks)****Marks**

- (a) Solve the equation  $2\log_e(x-5) = \log_e(2x-7)$  **3**

$$(x-5)^2 = 2x-7$$

$$x^2 - 10x + 25 = 2x - 7$$

$$x^2 - 12x + 32 = 0$$

$$(x-4)(x-8) = 0$$

$$x = 4, 8$$

As  $x-5 > 0$ ,  $x=8$  is the only solution.

- (b) Differentiate with respect to  $x$ :

i.  $y = e^{7x}$

$$y' = 7e^{7x}$$

ii.  $y = \frac{e^x}{x^2}$

$$y' = \frac{e^x \cdot x^2 - 2xe^x}{x^4}$$

$$= \frac{xe^x - 2e^x}{x^3}$$

$$= \frac{e^x(x-2)}{x^3}$$

- (c) Find the equation of the normal to the curve  $y = e^x(x^2 + 1)$  at  $(0, 1)$ . **3**

$$y' = e^x(2x) + e^x(x^2 + 1)$$

$$= e^x(x^2 + 2x + 1)$$

$$= e^x(x+1)^2$$

$$\text{At } (0, 1), m_{\text{tan}} = 1, \therefore m_{\text{nor}} = -1$$

$\therefore$  Eqn of normal is

$$y - 1 = -x$$

$$y = -x + 1$$

- (d) i. Solve for  $x$ :  $x^2 + 2x + 1 = 0$ . 1

$$(x+1)^2 = 0$$

$$x = -1$$

- ii. The displacement of an object at time  $(t)$  seconds is given by 3

$$x = e^{-2t} + 4e^{-t} - 2t$$

show that ~~find the time(s)~~ never

Hence, or otherwise, ~~find the time(s)~~ the object comes to rest. (Hint: Let  $m = e^{-t}$ )

$$\dot{x} = -2e^{-2t} - 4e^{-t} - 2$$

$$= -2(e^{-2t} + 2e^{-t} + 1)$$

$$= -2(m^2 + 2m + 1)$$

$$= -2(m+1)^2$$

$$\text{when at rest, } m = -1$$

$$\therefore e^{-t} = -1$$

no solutions

$\therefore$  never come to rest.

**End of paper.**

## Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

NESA STUDENT #: .....

Class (please ✓)

### Directions for multiple choice answers

- Read each question and its suggested answers.
- Select the alternative (A), (B), (C), or (D) that best answers the question.
- Mark only one circle per question. There is only *one* correct choice per question.
- Fill in the response circle completely, using blue or black pen, e.g.

(A) (B) ● (D)

- If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) (B) ~~●~~ ●

- If you continue to change your mind, write the word **correct** and clearly indicate your final choice with an arrow as shown below:

(A) (B) ~~●~~ ~~●~~ <sup>correct</sup>

1 — (A) (B) (C) (D)

2 — (A) (B) (C) (D)

3 — (A) (B) (C) (D)

4 — (A) (B) (C) (D)

5 — (A) (B) (C) (D)

6 — (A) (B) (C) (D)

7 — (A) (B) (C) (D)

8 — (A) (B) (C) (D)

9 — (A) (B) (C) (D)

10 — (A) (B) (C) (D)